

Diptadeep Roy

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EDUCATION

Adamas University

Master of Computer Applications (MCA)

Kolkata, India

2025 – 2027

Adamas University

Bachelor of Computer Applications (BCA)

Kolkata, India

2022 – 2025

CGPA: 6.65 / 10

TECHNICAL SKILLS

- **Programming Languages:** Python
- **Web & Backend:** HTML, Tailwind CSS, JavaScript, Flask
- **Blockchain:** Solidity, Geth, Clef
- **Databases:** MongoDB Atlas, MySQL
- **Core Concepts:** Data Structures & Algorithms (DSA), Database Management Systems (DBMS)

PROJECTS

Blockchain-Based Tokenization and Management of Land Documents

Sep 2024 – Apr 2025

Blockchain-based DApp for secure land-document tokenization and ownership management

Tech: Solidity, Flask, Python, IPFS, MongoDB, Geth, Clef, Remix

- Developed a blockchain-based decentralized application (DApp) for tokenizing land documents and representing ownership using ERC-721 non-fungible tokens.
- Integrated IPFS for decentralized document storage and used cryptographic content identifiers to provide tamper-evident document verification.
- Developed Flask-based backend services to connect the web application with blockchain smart contracts and MongoDB metadata storage.
- Designed and tested Solidity smart contracts using OpenZeppelin standards; deployed on a local private Ethereum network using Geth to achieve zero gas fees for testing and optimal throughput.
- Implemented secure account and private-key management using Clef while testing contract interactions through Remix IDE and Web3 interfaces.

PUBLICATIONS

- **Diptadeep Roy**, et al., “Blockchain-Based Tokenization and Management of Land Documents: A Decentralized Approach for Secure Ownership and Transfer,” *IEEE RACS 2025 / IEEE Xplore Digital Library*, 2025.
IEEE Xplore: ieeexplore.ieee.org/document/11413273

ACHIEVEMENTS

- Author and Presenter of IEEE-published research on blockchain-based land tokenization at IEEE RACS 2025.